

Audio Buffer and Line Driver with Low Quiescent Current

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Some of the pick-ups for guitars and other string instruments provide high output voltages. These require buffering before sending signals to the next stage of sound

PARTS LIST

Semiconductors:

- IC1 - TL064 quadruple operational amplifier
D1-D2 - 1N4148 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R2 - 10-kilo-ohm
R3-R4, R12 - 47-kilo-ohm
R5 - 1-mega-ohm
R6-R9 - 100-ohm
R10-R11 - 200-ohm

Capacitors:

- C1 - 150pF, ceramic disk
C2 - 0.1 μ F, ceramic disk
C3, C6 - 47 μ F, 35V electrolytic
C4 - 0.33 μ F, ceramic disk
C5 - 220 μ F, 35V electrolytic

Miscellaneous:

- CON1-CON4 - 2-pin connector
- 2 $\times$ 9V 6F22 or PP3 battery

processing equipment. You can also drive 600-ohm audio lines or high-impedance 600-4000-ohm headphones for self practicing with the instrument. Only several milliwatts of power is needed to drive these headphones, and that power can be provided by most operational amplifiers (op-amps).

In practice, when the buffer is powered by small dry batteries, like 6F22/PP3 for 9V, it is important to use buffer with low quiescent current. Quadruple operational ampli-

fier, like TL064, offers high input resistance and typical quiescent current of 0.8mA for all four internal op-amps in the package. Fortunately, op-amps in the package can be connected in parallel to increase the output drive capability necessary for driving 600-ohm loads.

Circuit and working

The circuit diagram of the simple buffer for guitar and line driver with low quiescent current is shown in Fig. 1.

The circuit is built around a low-power amplifier with JFET inputs TL064. The circuit can work with almost all quadruple operational amplifiers, including LM324, but the quiescent current and other parameters will be different.

The input resistance is 1Mega-ohm and depends mainly on resistor R5. Resistors R3, R4 and C3 are providing half of the Vcc (power supply) required to bias the output of the buffer. At test point TP3, we should obtain voltage level around Vcc/2.

Four op-amps are working as voltage followers. These are paralleled and resistors R6, R7, R8 and R9 are for equalisation of output currents. These resistors are important to prevent activation of the current-limiting function of op-amps and not to introduce additional distortions.

In order to keep the buffer simple, no potentiometers are used. The volume and tone are controlled with potentiometers on the musical instrument or on the next stage of the sound processing equipment.

Connectors CON3 and CON4 are used for connecting the loads. If one of the connectors is used for 600-ohm load, the second connector should be used for load with

Test Points

Test point	Details
TP0	0V (GND)
TP1	18V
TP2	Guitar input signal
TP3	Around 9V

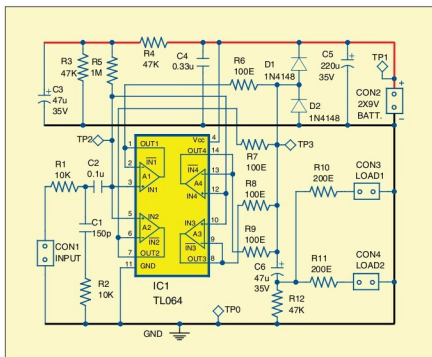


Fig. 1: Circuit diagram of the audio buffer and line driver

